

UNIT I: ARTIFICIAL NEURAL NETWORKS

1. **Define a biological neuron.** A biological neuron is the fundamental processing unit of the brain. It consists of three main parts: **Dendrites** (receive signals), the **Cell Body** (processes signals), and the **Axon** (transmits signals).
2. **What is the McCulloch-Pitts neuron model?** It is the earliest mathematical model of a biological neuron where binary inputs are weighted and summed; if the sum exceeds a threshold, the neuron fires an output.
3. **Differentiate between linear and non-linear activation functions.** Linear functions produce an output proportional to the input, while non-linear functions (like Sigmoid or ReLU) allow the network to learn complex, non-linear mappings between inputs and outputs.
4. **Define a Single Layer Perceptron.** It is the simplest form of a neural network used for classifying linearly separable data, consisting of a single layer of weights and a threshold activation function.
5. **What is a "bias" in a neuron and why is it needed?** A bias is an additional parameter added to the weighted sum of inputs. It allows the activation function to be shifted left or right, enabling the model to better fit the data when all input values are zero.
6. **Define the Learning Rule in a Perceptron.** The learning rule is a procedure for modifying the weights and biases of a network to minimize the difference between the actual output and the desired target.
7. **Explain the term "Linearly Separable."** Data is linearly separable if a single straight line (in 2D) or a hyperplane (in higher dimensions) can completely separate two different classes.

UNIT II: BACKPROPAGATION NEURAL NETWORKS

1. **Define a Multi-Layer Perceptron (MLP).** An MLP is a feedforward network with at least one hidden layer between the input and output layers, capable of solving non-linearly separable problems.
2. **Explain how local minima can affect the accuracy of training.** During gradient descent, the model may get stuck in a sub-optimal point (local minimum) where the error is low but not the absolute lowest, preventing the network from reaching peak accuracy.
3. **What is "Network Paralysis"?** This occurs when weights become very large, causing activation functions to saturate (slope becomes zero), which stops the learning process as no gradient flows back.
4. **Define Retrained Deep Neural Networks.** These are pre-trained models that are updated with new data for a specific task to enhance performance and stability without training from scratch.
5. **What is "Temporal Instability" in training?** It refers to a situation where the network's performance fluctuates wildly during training, often due to a high learning rate or the vanishing/exploding gradient problem.
6. **Define the "Forward Pass" and "Backward Pass."** In the **Forward Pass**, input data travels through the layers to generate a prediction. In the **Backward Pass**, the error is calculated and propagated back through the network to update the weights.

7. **Mention the "Practical Considerations" of Backpropagation.** Key considerations include selecting an appropriate initial weight distribution, choosing a loss function, and determining the optimal learning rate.

UNIT III: CONVOLUTIONAL NETWORKS AND SEQUENCE MODELLING

1. **Describe how the pooling layer helps in reducing computation.** Pooling (like Max Pooling) down-samples the input by reducing its spatial dimensions, which lowers the number of parameters and computational cost.
2. **What is a Bidirectional RNN?** A BRNN processes sequence data in both directions (forward and backward) to capture context from both the past and the future simultaneously.
3. **Define Auto-regressive networks.** These are networks where the output at a specific time step is modeled as a function of its own previous outputs.
4. **What are "Long-term dependencies"?** They refer to the relationship between elements in a sequence that are separated by many time steps, which standard RNNs often struggle to capture due to vanishing gradients.
5. **What is a "Kernel" (or Filter) in CNN?** A kernel is a small matrix of weights that slides over the input image to perform the convolution operation, extracting specific features like edges or textures.
6. **Define "Recursive Neural Networks."** Unlike standard RNNs that process sequences linearly, Recursive Neural Networks apply the same set of weights over a hierarchical or tree-like structure to process data.
7. **What is "Padding" and why is it used in CNNs?** Padding involves adding extra pixels (usually zeros) around the border of an image before convolution. It is used to maintain the spatial dimensions of the output and prevent information loss at the edges.
8. **What are 'Temporal Dependencies' in sequence modeling?** Temporal dependencies refer to the relationship or correlation between data points at different points in time within a sequence. Modeling these dependencies allows the network to understand how an earlier event influences a later outcome.
9. **What is the 'Vanishing Gradient Problem' in long-term dependencies?** Standard RNNs struggle to learn long-term dependencies because, during backpropagation through many time steps, the gradients can become extremely small (vanish). This prevents the weights from updating effectively, making the network "forget" information from distant past time steps.
10. **How do LSTM or GRU networks address long-term dependencies?** Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU) use specialized "gates" (input, forget, and output gates) to regulate the flow of information. These gates allow the network to selectively remember or discard information over long intervals, effectively overcoming the vanishing gradient problem.

UNIT IV: PROBABILISTIC MODELS AND AUTOENCODERS

1. **What is meant by a structured probabilistic model?** It is a framework that uses graphs (nodes and edges) to represent and factorize the probability distribution of multiple random variables.
2. **Define Latent Variables in Linear Factor Models.** Latent variables are hidden factors that are not directly observed but influence the observed data; factor models seek to extract these to understand data relationships.
3. **Compare Monte Carlo models and linear factor models.** Monte Carlo models use random sampling for inference in complex systems, while linear factor models identify hidden structures in observed data.
4. **Define an Autoencoder.** An Autoencoder is an unsupervised network that learns to compress input data into a code (encoding) and then reconstruct it (decoding) with minimal error.
5. **Explain "Dimensionality Reduction" in the context of Autoencoders.** Autoencoders achieve dimensionality reduction by forcing the input into a smaller hidden layer (the bottleneck), which acts as a compressed representation containing only the most vital features.
6. **What is "Gibbs Sampling"?** It is a Markov Chain Monte Carlo (MCMC) algorithm used to generate a sequence of samples from the joint probability distribution of two or more random variables when direct sampling is difficult.
7. **Define a "Markov Network" (Undirected Graph).** A Markov network represents dependencies using undirected edges, where the lack of an edge between two nodes implies that they are conditionally independent given the rest of the variables.
8. **Define an Autoencoder.** An Autoencoder is an unsupervised artificial neural network used to learn efficient data codings. It consists of two main parts: an **encoder** that compresses the input into a latent-space representation, and a **decoder** that reconstructs the input from this representation.
9. **What is the "Bottleneck" in an Autoencoder?** The bottleneck is the hidden layer with the smallest number of neurons in the architecture. It forces the network to learn a compressed, "lossy" representation of the input data, effectively capturing only the most important features.
10. **What is meant by "Reconstruction Error"?** Reconstruction error is a loss function (such as Mean Squared Error) that measures the difference between the original input and the output generated by the decoder. The network is trained to minimize this error to ensure high-quality data reproduction.
11. **Define Denoising Autoencoders.** Denoising Autoencoders are a variant designed to learn robust features by being trained to reconstruct a "clean" version of an input from a "corrupted" or noisy version.
12. **State the primary application of Autoencoders.** Autoencoders are primarily used for **dimensionality reduction**, feature learning, and data denoising.
13. **What is a Structured Probabilistic Model?** It is a framework that uses graphs to represent a probability distribution over a set of random variables. These models describe which variables depend on each other directly, helping to simplify complex multi-variable systems.

14. **Define a Directed Graphical Model (Bayesian Network).** A directed graphical model uses a Directed Acyclic Graph (DAG) where nodes represent random variables and directed edges represent causal or conditional dependencies.
15. **What is an Undirected Graphical Model (Markov Network)?** It is a graph where edges have no direction, representing symmetric dependencies between variables. It is used when relationships are mutual rather than causal.
16. **What are "Latent Variables" in probabilistic modeling?** Latent variables are "hidden" variables that are not directly observed in the data but are inferred from observed variables to explain underlying patterns.
17. **Define the "Inference" problem in probabilistic models.** Inference is the process of using the model to compute the probability distribution over a subset of variables, given that the values of other variables have been observed.
18. **What are Linear Factor Models?** These are models that represent the observed data as a linear combination of a small number of independent latent factors plus noise. They are used to identify the simplest underlying structure of the data

UNIT V: ACCELERATING DEEP LEARNING MODELS

1. **Define ImageNet.** ImageNet is a large-scale visual database used for training and benchmarking deep learning models in image recognition.
2. **Name two techniques to accelerate deep learning models. Model Pruning** (removing redundant weights) and **Model Compression** (quantization) are used to make models faster for deployment.
3. **What is a Generative Adversarial Network (GAN)?** A GAN consists of two networks—a Generator and a Discriminator—that compete to produce realistic synthetic data.
4. **What is the significance of ResNet?** ResNet introduced "Residual Blocks" (skip connections), allowing for the training of extremely deep networks by overcoming the vanishing gradient problem.
5. **What is a "Residual Block"?** Introduced in ResNet, a residual block uses "skip connections" to pass the input of a layer directly to the output of a later layer, helping to train much deeper networks.
6. **Define "Model Pruning."** Pruning is the process of removing redundant or less significant neurons/weights from a trained model to make it smaller and faster without a major loss in accuracy.
7. **What is a "Quantum GAN" (QGAN)?** A QGAN is a generative model that utilizes quantum circuits for the generator or discriminator (or both), potentially offering speedups in learning complex data distributions.